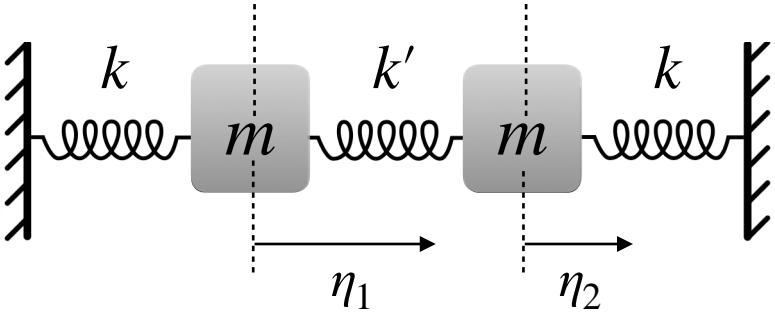
PROBLEM SET 6

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 100)

Fetter & Walecka, Ch. 4

1) **(25 pts)** Reconsider the coupled oscillator example covered in class lecture.



a) **(2 pts)** Write the Lagrange equations of motions for this system

b) **(3 pts)** Try a real normal-mode solution of the type:

$$\eta_\lambda = C\rho_\lambda \cos(\omega t + \phi)$$

where (λ) represents the generalized coordinate index, ρ_λ represents the amplitude (eigenvector), ω is a normal-mode frequency, and (C, ϕ) represent overall constants determined from the initial conditions, $\eta(t = 0)$, $\dot{\eta}(t = 0)$, if they are given.

Substitute this solution to the coupled equations in part (a) and re-write the system of linear equations in matrix form:

$$(v_{\sigma\lambda} - \omega^2 m_{\sigma\lambda})\rho_\lambda = 0,$$

and explicitly identify the potential energy $(v_{\sigma\lambda})$ and mass $(m_{\sigma\lambda})$ matrices, as they will be used in the later parts.

To solve the homogeneous matrix equation in part (b), for a non-trivial solution $\rho_\lambda \neq 0$, requires $\det|\, v_{\sigma\lambda} - \omega^2 m_{\sigma\lambda}\,| = 0$, which yields a **characteristic** or **secular** equation of degree n (number of generalized coordinates, in this case is 2) in ω^2, meaning there are two eigenfrequencies or modes of oscillations. Determine each of these eigenfrequencies, ω_s^2. Define the notation for each mode: $\omega_-^2 \equiv \omega_1^2, \omega_+^2 \equiv \omega_2^2$

c) **(5 pts)** Determine the relative amplitudes, $\rho_\lambda^{(s)}$, for each mode s, to show they can be expressed as:

$$\rho_1^{(1)} = \rho_2^{(1)}, \quad \rho_1^{(2)} = -\rho_2^{(2)}$$

Hint: Substitute the eigenfrequency, ω_s^2, back in the system of linear equations determined in part (b). Note that the system of n linear equations is **NOT** linearly independent, and therefore, either equation will give a ratio of the amplitudes, $\rho_1^{(s)}/\rho_2^{(s)}$ for each particular mode s.

NOTE: the ratio of amplitudes describes the relative motion of the blocks for a particular mode. For example, it might tell us whether the blocks are moving together in the same direction or in opposite direction.

d) **(5 pts)** Although not usually necessary to find the normalization factor in general, it is asked here for completion. Use the orthonormality condition, $\vec{\rho}^{(t)T}\mathbf{M}\vec{\rho}^{(s)} = \mathbf{I}$ or in index notation: $\sum_\lambda \sum_\sigma \rho_\sigma^{(t)} m_{\sigma\lambda} \rho_\lambda^{(s)} = \delta_{st}$ to calculate the normalization constant for each normal mode eigenvector, and show that the eigenvectors can be expressed as:

$$\vec{\rho}^{(1)} = \frac{1}{\sqrt{2m}}\begin{bmatrix}1\\1\end{bmatrix}, \quad \vec{\rho}^{(2)} = \frac{1}{\sqrt{2m}}\begin{bmatrix}-1\\1\end{bmatrix}$$

Hint: Insert the results of part (d) into the eigenvector: $\vec{\rho}^{(s)} = \begin{bmatrix}\rho_1^{(s)}\\ \rho_2^{(s)}\end{bmatrix}$, and along with the results of part (b), use the normalization condition, $\delta_{st} = 1,\ t = s$ to determine the normalization factor.

e) **(5 pts)** Construct the modal matrix (first read **Background)**,

$$\mathbf{A} = \begin{bmatrix} \rho_1^{(1)} & \rho_1^{(2)} \cdots \rho_1^{(n)} \\ \cdots\cdots\cdots\cdots \\ \rho_n^{(1)} & \rho_n^{(2)} \cdots \rho_n^{(n)} \end{bmatrix} = \left[\vec{\rho}^{(1)}, \vec{\rho}^{(2)}, \ldots \vec{\rho}^{(n)}\right]$$

from the generic column eigenvectors, (do not include the normalization factors from (e), but rather use the generic form) and use the relation, $\vec{\eta} = \mathbf{A}\vec{\zeta}$, along with the relative amplitudes relations [part (d)] to show that the generalized coordinates can be expressed in terms of the normal coordinates as a linear combination of the normal mode coordinates with the eigenvector coefficients

$$\eta_1 = \rho_1^{(1)}\zeta_1 - \rho_2^{(2)}\zeta_2, \quad \eta_2 = \rho_1^{(1)}\zeta_1 + \rho_2^{(2)}\zeta_2$$

Then solve for (ζ_1, ζ_2) above to construct the normal-mode coordinates (ζ) in terms of the generalized coordinates to show that they have the generic form:

$$\zeta_1 = \frac{\eta_1 + \eta_2}{2\rho_1^{(1)}}, \quad \zeta_2 = \frac{\eta_2 - \eta_1}{2\rho_2^{(2)}}$$

Finally, use the normalization result from part (e) to show that the normal-mode coordinates take the form:

$$\begin{bmatrix} \zeta_1 \\ \zeta_2 \end{bmatrix} = \sqrt{\frac{m}{2}} \begin{bmatrix} \eta_2 + \eta_1 \\ \eta_2 - \eta_1 \end{bmatrix}$$

Background [see Fetter & Walecka Ch. 4 p.98-99]**:** The modal matrix is square matrix constructed from the column eigenvectors, $\mathbf{A} \equiv \left[\vec{\rho}^{(1)}, \vec{\rho}^{(2)}, \ldots \vec{\rho}^{(n)}\right]$, where the superscript represents the normal mode. It has the property of simultaneously *diagonalizing* the mass and potential energy matrices, as follows: $\mathbf{A}^T\mathbf{M}\mathbf{A} = \mathbf{I}$,

$$\mathbf{A}^T\mathbf{V}\mathbf{A} = \begin{bmatrix} \omega_1^2 & 0 \\ 0 & \omega_2^2 \end{bmatrix} \equiv \omega_D^2$$

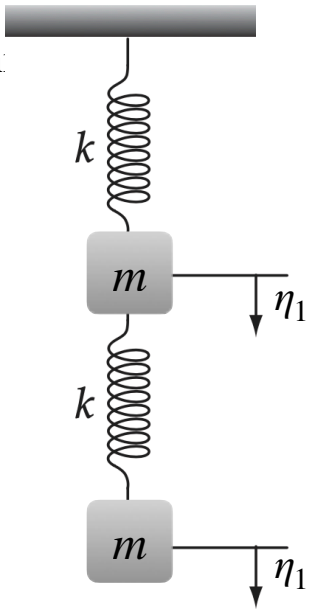
The modal matrix also relates the original generalized coordinates, η_σ, to a new set of coordinates, ζ_λ, (normal-mode coordinates) thru the relation: $\eta_\sigma = \sum_\lambda A_{\sigma\lambda}\zeta_\lambda$, or in matrix notation $\vec{\eta} = \mathbf{A}\vec{\zeta}$

Furthermore, the normal coordinates relation to generalized coordinates can be obtained by making use of the diagonalizing property of the mass matrix, $(\mathbf{A}^T\mathbf{M})\vec{\eta} = \underbrace{\mathbf{A}^T\mathbf{M}\mathbf{A}}_{\mathbf{I}}\vec{\zeta}$ to obtain:

$\vec{\zeta} = (\mathbf{A}^T\mathbf{M})\vec{\eta}$, or in the more familiar notation: $\begin{bmatrix} \zeta_1 \\ \zeta_2 \end{bmatrix} = \mathbf{A}^T\mathbf{M}\begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix}$, where it is important to note that the index $\zeta_{1,2}$, represent the s-normal mode, whereas the index $\eta_{1,2}$, represents the generalized coordinate, e.g. in this example, describing the displacement of each block. In effect, each normal-mode coordinate is linear combination of the generalized coordinates.

f) **(5 pts)** Verify that the equations of motion in normal-mode coordinates are de-coupled, and thus have the form of a simple harmonic oscillator with frequency of the corresponding normal mode.
Hint: to verify this, take two time derivatives of each normal mode coordinate and show that: $\ddot{\zeta}_s = -\omega_s^2\zeta,\ s = 1,2$

2) **(25 pts)** A particle of mass m is attached to a rigid support by a spring with force constant k. At equilibrium, the spring hangs vertically downward. To this mass-spring combination is attached an identical oscillator, the spring of the latter being connected to the mass of the former.



a) **(5 pts)** Write the Lagrange equations of motion for th system.

b) **(10 pts)** Try a real normal-mode solution of the type:

$$\eta_\lambda = C\rho_\lambda \cos(\omega t + \phi)$$

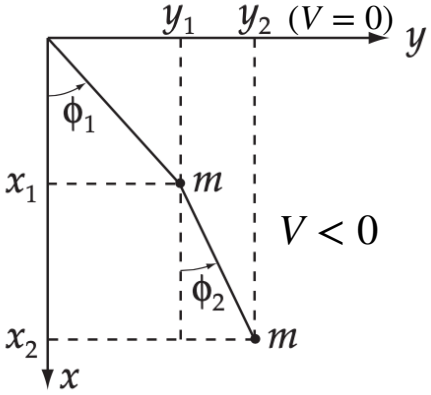
into the Lagrange equations of motion and

(i) identify the potential energy ($v_{\sigma\lambda}$) and mass ($m_{\sigma\lambda}$) matrices

(ii) solve for the eigenfrequencies, ω_s^2, of each normal mode s to show they can be expressed as: $\omega_1^2 = \dfrac{3+\sqrt{5}}{2}\dfrac{k}{m}$, $\omega_2^2 = \dfrac{3-\sqrt{5}}{2}\dfrac{k}{m}$

c) **(10 pts)** Determine the relative amplitudes, $\rho_\lambda^{(s)}$, for each mode s to show they can be expressed as: $\dfrac{1-\sqrt{5}}{2}\rho_1^{(1)} = \rho_2^{(1)}$, $\dfrac{1+\sqrt{5}}{2}\rho_1^{(2)} = \rho_2^{(2)}$

3) **(25 pts)** A simple pendulum consists of a bob of mass m suspended by an inextensible (and massless) string of length ℓ. From the bob of this pendulum is suspended a second, identical pendulum. Consider the case of small oscillations ($\sin\phi \approx \phi, \cos\phi \approx 1 - \phi^2/2$)



a) **(5 pts)** Calculate the Lagrangian, $L = T -$ small angle oscillations, where

$T = \frac{1}{2} m\ell^2 (2\dot{\phi}_1^2 + 2\dot{\phi}_1\dot{\phi}_2 + \dot{\phi}_2^2)$ and

$V = V_0 + \frac{1}{2} mg\ell(2\phi_1^2 + \phi_2^2)$, where

$V_0 = -3mg\ell$

b) **(5 pts)** Write the Lagrange equations of motion for (ϕ_1, ϕ_2) for this system using a small angle approximation. (From now on, let $\phi_\lambda \to \eta_\lambda$, for small displacements)

c) **(5 pts)** Try a real normal-mode solution of the type:

$$\eta_\lambda = C\rho_\lambda \cos(\omega t + \theta)$$

into the Lagrange equations of motion and

(i) identify the potential energy ($v_{\sigma\lambda}$) and mass ($m_{\sigma\lambda}$) matrices and show they take the form:

$$m_{\sigma\lambda} = m\ell^2 \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad v_{\sigma\lambda} = mg\ell \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

(ii) solve for the eigenfrequencies, ω_s^2 of each normal mode s to show they can be expressed as

$$\omega_1^2 = \left(2 + \sqrt{2}\right)\frac{g}{\ell}, \quad \omega_2^2 = \left(2 - \sqrt{2}\right)\frac{g}{\ell}$$

d) **(5 pts)** Determine the relative amplitudes, $\rho_{\lambda}^{(s)}$, for each mode s to show they can be expressed as:

$$\rho_2^{(1)} = -\sqrt{2}\rho_1^{(1)}, \quad \rho_2^{(2)} = \sqrt{2}\rho_1^{(2)}$$

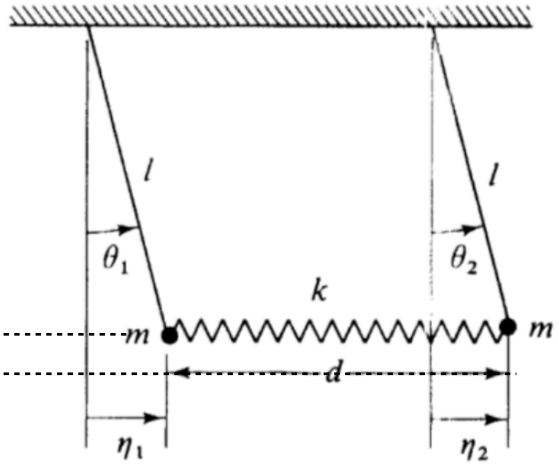
e) **(5 pts)** Construct the modal matrix, $\mathbf{A}$, and use the relation, $\vec{\eta} = \mathbf{A}\vec{\zeta}$, to show that the generalized coordinates can be expressed in terms of the normal coordinates as a linear combination of the normal mode coordinates with the eigenvector coefficients

$$\eta_1 = \rho_1^{(1)}\zeta_1 + \frac{1}{\sqrt{2}}\rho_2^{(2)}\zeta_2, \quad \eta_2 = -\sqrt{2}\rho_1^{(1)}\zeta_1 + \rho_2^{(2)}\zeta_2$$

then solve for the normal coordinates, ζ_1, ζ_2 to show:

$$\zeta_1 = \frac{\sqrt{2}\eta_1 - \eta_2}{2\sqrt{2}\rho_1^{(1)}}, \qquad \zeta_2 = \frac{\sqrt{2}\eta_1 + \eta_2}{2\rho_2^{(2)}}$$

4) **(25 pts)** Two pendula of equal length l and equal mass m are connected by a spring of force constant k. Consider the case of small oscillations ($\sin\theta \approx \theta, \cos\theta \approx 1 - \theta^2/2$), where $\eta = l\sin\theta \approx l\theta$ [see Fetter & Walecka, Ch. 4 example]



a) **(5 pts)** Calculate the Lagrangian, $L = T -$ for small angle oscillations

b) **(5 pts)** Write the Lagrange equations of m for (η_1, η_2) for this system using a small angle approximation.

c) **(5 pts)** Try a real normal-mode solution of the type:

$$\eta_\lambda = C\rho_\lambda \cos(\omega t + \phi)$$

into the Lagrange equations of motion and

(i) identify the potential energy ($v_{\sigma\lambda}$) and mass ($m_{\sigma\lambda}$) matrices and show they take the form:

$$m_{\sigma\lambda} = m\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad v_{\sigma\lambda} = \begin{bmatrix} \frac{mg}{l} + k & -k \\ -k & \frac{mg}{l} + k \end{bmatrix}$$

(ii) solve for the eigenfrequencies, ω_s^2 of each normal mode s to show they can be expressed

as: $\omega_1^2 = \dfrac{g}{l}, \quad \omega_2^2 = \dfrac{g}{l} + 2\dfrac{k}{m}$

d) **(5 pts)** Determine the relative amplitudes, $\rho_\lambda^{(s)}$, for each mode s to show they can be expressed as:
$\rho_1^{(1)} = \rho_2^{(1)}, \;\; \rho_1^{(2)} = -\rho_2^{(2)}$

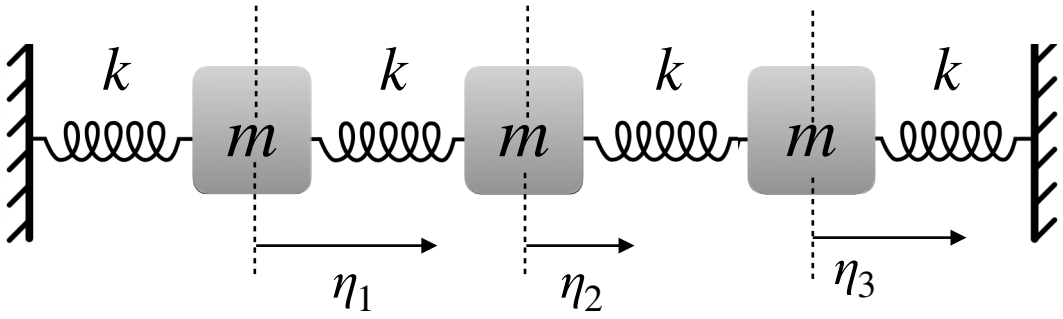
Use the normalization condition, $\vec{\rho}^{(t)T}\mathbf{M}\vec{\rho}^{(s)} = \mathbf{I}$, to show the normalization is $\rho_1^{(1)} = 1/\sqrt{2m}, \;\; \rho_2^{(2)} = 1/\sqrt{2m}$

Hint: Using the relative amplitudes, note you can express the eigenvectors as: $\vec{\rho}^{(1)} = \begin{bmatrix} \rho_1^{(1)} \\ \rho_2^{(1)} \end{bmatrix} = \rho_1^{(1)} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, and similarly for $\vec{\rho}^{(2)} = \begin{bmatrix} \rho_1^{(2)} \\ \rho_2^{(2)} \end{bmatrix} = \rho_2^{(2)} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

e) **(5 pts)** Construct the modal matrix, $\mathbf{A}$, and use the relation, $\vec{\zeta} = (\mathbf{A}^T\mathbf{M})\vec{\eta}$, to show that the normal-mode coordinates (ζ) in terms of the generalized coordinates have the form:

$$\begin{bmatrix} \zeta_1 \\ \zeta_2 \end{bmatrix} = \sqrt{\frac{m}{2}} \begin{bmatrix} \eta_1 + \eta_2 \\ \eta_1 - \eta_2 \end{bmatrix}$$

5) **(+10 pt extra credit Exam II)** Reconsider the coupled oscillator example covered in class lecture, but add an additional block + spring.



a) Calculate the Lagrangian, $L = T - V$

b) Write the Lagrange equations of motion for (η_1, η_2, η_3) for this system

c) Try a real normal-mode solution of the type:

$$\eta_\lambda = C\rho_\lambda \cos(\omega t + \phi)$$

into the Lagrange equations of motion and

(i) identify the potential energy $(v_{\sigma\lambda})$ and mass $(m_{\sigma\lambda})$ matrices and show they take the form:

$$m_{\sigma\lambda} = m \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad v_{\sigma\lambda} = \begin{bmatrix} 2k & -k & 0 \\ -k & 2k & -k \\ 0 & -k & 2k \end{bmatrix}$$

(ii) solve for the eigenfrequencies, ω_s^2 of each normal mode s to show they can be expressed

as: $\omega_1^2 = \dfrac{2k}{m}, \quad \omega_2^2 = (2+\sqrt{2})\dfrac{k}{m}, \quad \omega_3^2 = (2-\sqrt{2})\dfrac{k}{m}$

d) Determine the relative amplitudes, $\rho_\lambda^{(s)}$, for each mode s to show they can be expressed as:

$$\rho_1^{(1)} = -\rho_3^{(1)}, \quad \rho_2^{(1)} = 0$$

$$\rho_1^{(2)} = \rho_3^{(2)}, \qquad \rho_2^{(2)} = -\sqrt{2}\rho_3^{(2)}$$

$$\rho_1^{(3)} = \rho_3^{(3)}, \qquad \rho_2^{(3)} = \sqrt{2}\rho_3^{(3)}$$

e) Construct the modal matrix, $\mathbf{A}$. Then use the relation, $\vec{\eta} = \mathbf{A}\vec{\zeta}$, along with the relative amplitudes [part (d)] to show that the generalized coordinates can be expressed in terms of the normal coordinates as a linear combination of the normal mode coordinates with the eigenvector coefficients

$$\eta_1 = \rho_1^{(1)}\zeta_1 - \frac{1}{\sqrt{2}}\rho_2^{(2)}\zeta_2 + \rho_3^{(3)}\zeta_3$$

$$\eta_2 = \rho_2^{(2)}\zeta_2 + \sqrt{2}\rho_3^{(3)}\zeta_3$$

$$\eta_3 = -\rho_1^{(1)}\zeta_1 - \frac{1}{\sqrt{2}}\rho_2^{(2)}\zeta_2 + \rho_3^{(3)}\zeta_3$$

then solve for the normal coordinates, $(\zeta_1, \zeta_2, \zeta_3)$ to show:

$$\zeta_1 = \frac{\eta_1 - \eta_3}{2\rho_1^{(1)}}, \quad \zeta_2 = \frac{-\eta_1 + \sqrt{2}\eta_2 - \eta_3}{2\sqrt{2}\rho_2^{(2)}}, \quad \zeta_3 = \frac{\eta_1 + \sqrt{2}\eta_2 + \eta_3}{4\rho_3^{(3)}}$$